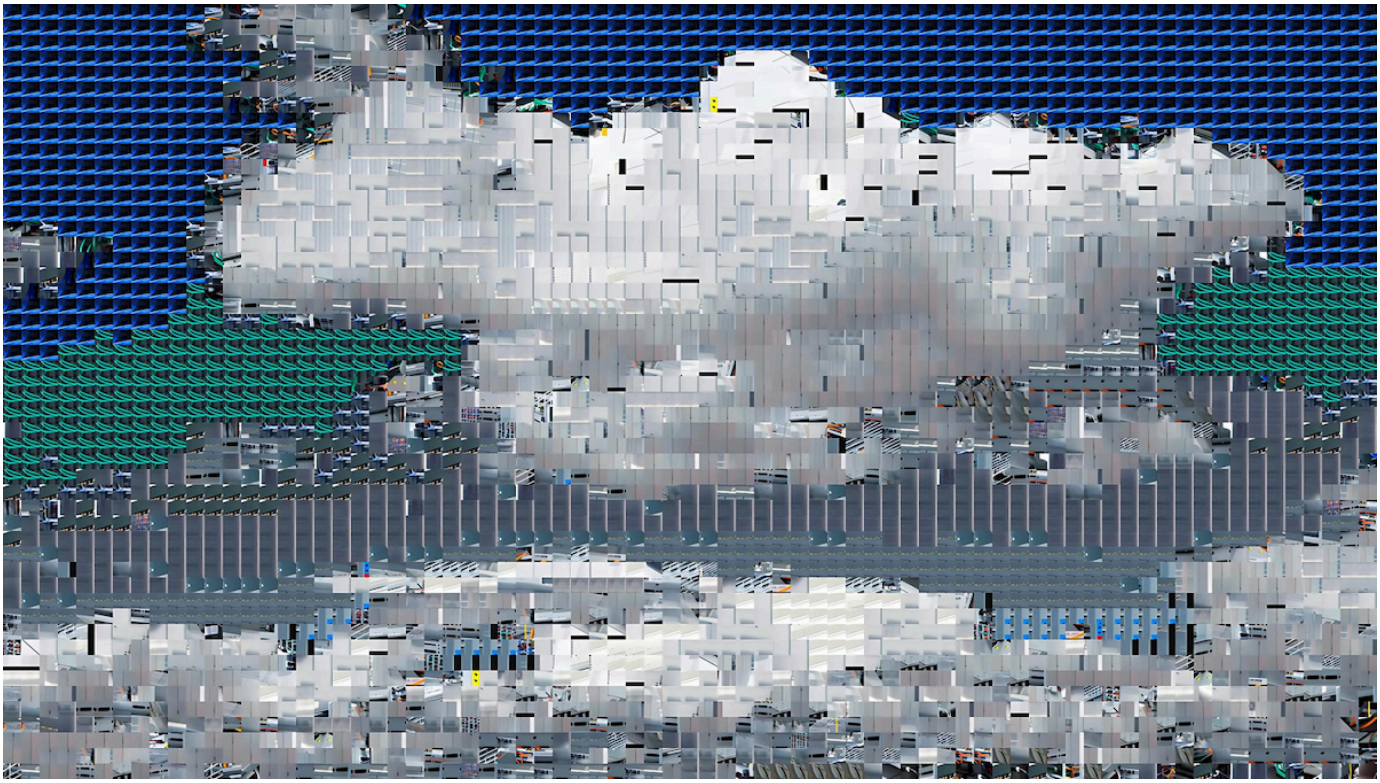
PERSPECTIVE

Clouds Over Public Infrastructure: Rethinking Internet Governance in the Hyperscaler Era

CORINNE CATH / APR 17, 2025



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Cloud computing—the often invisible backbone of our digital environment—has suddenly become a hot-button political issue. In the wake of Trump's November 2024 election victory, governments and institutions worldwide are reassessing their digital dependencies on American tech giants. This shift marks a growing awareness that cloud dependencies can create new vulnerabilities during times of geopolitical instability. Around the globe, governments, industries, and individuals now face a sobering reality: their digital infrastructure—and everything that relies on it—could be subject to sudden policy shifts, tariffs, or even access restrictions if tensions with the United States escalate.

In February 2024, I, along with cybersecurity expert Bert Hubert, raised concerns about the Stichting Internet Domeinregistratie Nederland (SIDN) decision to transfer part of its registry operations to Amazon Web Services (AWS). What began as a technical decision by a little-known Dutch internet governance organization evolved into a nationwide reckoning about something more fundamental: how cloud migration can transform public institutions from within.

This seemingly routine infrastructure choice sparked an unexpectedly profound debate about control, institutional independence, and the future of the internet. The controversy ultimately led the Dutch government to temporarily halt not only SIDN's move to AWS but also its plans to migrate its digital workspaces to the Microsoft cloud. That this escalation emerged from our initial challenge to SIDN shows that resistance to hyperscaler dominance is not merely theoretical—it is politically possible. This piece draws from my recent critical infrastructure lab report, which puts SIDN's AWS move into stark relief and explores how this case exemplifies broader tensions between public interest mandates and commercial cloud imperatives.

At its core, this case reveals a growing friction between the gravitational pull of the cloud and the need to preserve the public interest in digital infrastructure decisions. In this small, pragmatic nation of 18 million—where even the land was engineered from water—the Dutch recent experience with re-engineering its vital .nl Internet governance organization offers lessons and warnings for a global community reckoning with cloud dependency. It also shows that building a better conceptual narrative for this debate, and socio-economic frameworks for computing, are as crucial as, if not more so than, incremental resistance through bare-metal alternatives.

The SIDN-AWS controversy: more than digital sovereignty

In the Netherlands, SIDN manages the digital registry of over 6 million '.NL' websites, including critical government domains. In January 2024, after decades of operating its services in-house, SIDN announced plans to outsource part of its registry operations to Amazon Web Services (AWS). When the "Dutchness" of .nl appeared threatened by a commercial American corporation, the debate initially centered on digital sovereignty and autonomy. Headlines like "How Dutch is .NL?" and "Minister Puts Brakes on SIDN's Proposed Move to AWS" capture more than bureaucratic hesitation. Parliament members across the political spectrum raised formal questions, while the Ministry of Economic Affairs launched an investigation into risks to the Netherlands's "digital open strategic autonomy."

The ensuing political response has been wavering. In January 2025, the Dutch government reluctantly approved the migration, despite acknowledging "significant national security risks." The approval came with additional oversight mechanisms that many experts dismissed as "mustard after the meal"—a Dutch expression for measures that come too late to be effective. But by March 2025, sustained pressure from a coalition of political parties led the Dutch parliament to unanimously halt the migration, demanding renewed consultations between the Ministry of Economic Affairs, SIDN, and national cloud providers.

It specifically aimed to prevent any part of the Domain Name System (DNS) chain—the critical infrastructure that SIDN manages as the official registry for all .nl domain names, maintaining the authoritative database that ensures Dutch websites remain accessible—from moving to Amazon. The most recent political intervention only temporarily halts the move to the cloud. However, it is sparking deeper questions about cloud dependency across Dutch digital infrastructure, including the possibility of reversing the government's ongoing migration to Microsoft cloud services and developing local alternatives for universities and other public institutions.

Emerging alternatives: from critique to action?

The SIDN controversy is catalyzing a broader movement across Dutch society that is beginning to imagine and build alternative digital infrastructure. This resistance is emerging at multiple levels with experimental approaches to reducing hyperscaler dependency. The Dutch cloud community is transforming critique into concrete alternatives through initiatives like "Cloud Kootwijk", which brings together local providers to develop viable alternatives to hyperscaler clouds. As one entrepreneur bluntly stated, they are "putting their metal where their mouth is." While modest in scale compared to global giants, these collective efforts demonstrate that different technical architectures are possible in some use cases. This effort is commendable and necessary.

At the same time, its reference and deference to "Radio Kootwijk" raises concerns. It provides different possibilities. Yet, simultaneously, this effort overlooks the Dutch colonial past in ways that problematically romanticize the nation's imperial history—building on this legacy makes for shaky foundations of an emancipatory technological alternative. Alternatives for the corporate cloud require it to be built on a foundation that integrates post-colonial awareness with cutting-edge chips—combining 2025's technological capabilities with contemporary historical consciousness.

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These local industry-led initiatives are complemented by parallel efforts within the public sector, where local governments are developing their own approaches to address these challenges. At the municipal level, the City of Amsterdam is exploring what digital independence means and how to achieve it. Their report, "Striving for Digital Autonomy in Amsterdam," outlines the city's efforts to reduce its dependency on large tech companies for digital services and products. It outlines three strategies: strengthening European and national

collaboration, maintaining control over data, and developing digitally independent infrastructure and applications. The municipality acknowledges that complete independence is not achievable due to technical and financial constraints. Hence, the city is pursuing a step-by-step approach by incorporating "digital autonomy" into existing policies. It also aligns with European and national initiatives to gradually move away from Big Tech, creating replicable models for other public entities.

These debates echo across Europe. In February 2025, Francesca Bria, a fellow at Stiftung Mercator, and colleagues published the EuroStack concept, a conceptual blueprint for "European technological sovereignty." Bria argues that with over 80% of Europe's digital products and services being imported, the continent faces dependencies on Big Tech that threaten its autonomy and economic stability. The proposed EuroStack would span all technology layers—from hardware and cloud infrastructure to data management, AI applications, and governance frameworks. In the process, creating a European technological model prioritizes public interest and democratic values. Bria advocates for a €10 billion European Technological Sovereignty Fund to drive this initiative, alongside policy changes such as prioritizing European-native tech in public procurement, encouraging open-source adoption, and ensuring interoperability. The vision positions Europe as offering a democratic alternative to both "the extractive and monopolistic models of US Big Tech and the authoritarian Chinese Big State," creating what Bria calls a "European Big Democracy" approach to digital infrastructure.

The discussion on European alternatives to the corporate cloud also lives in the industry. In March 2025, 70 European companies, including Airbus and OVHcloud, sent an open letter to the EU Commission. They warn that Europe faces a critical digital dependency crisis, threatening its sovereignty and economic future. The signatories, united behind the "EuroStack Initiative," call for "radical action" by implementing "Buy European" procurement mandates, pooling existing European digital resources, prioritizing services with strong adoption potential, and developing sovereign cloud certification standards. The letter stresses that without immediate action, Europe's dependence on foreign technologies will be irreversible within three years.

These emerging alternatives, while promising, remain speculative and untested in real-world conditions. While these industry- and research-led proposals echo many of the valid concerns raised in the Dutch context, they also risk falling into sovereignty-oriented solutions that overlook deeper structural challenges. Genuine digital transformation demands more than replacing foreign technology with local equivalents or building new infrastructures on outdated colonial imaginings. It requires radically reimagining the underlying political economy that shapes how digital systems operate and who benefits from them. The current framing of the debate about whether the cloud is politically effective in mobilizing opposition. Yet, it inadvertently narrows the discussion to questions of national, or European, control and chasing European competitiveness, as others also argue, rather than addressing the more fundamental changes caused by commercial cloud imperatives.

Beyond sovereignty: the institutional changes at stake

The focus on American corporate control, sovereignty, and competition misses the deeper issue: how cloud migration reconfigures the very nature of public institutions, as leading thinkers argue. These issues cannot be addressed by increased competitiveness or national control. When organizations like SIDN migrate to hyperscaler clouds, they don't simply move their computing needs elsewhere. Rather, they undergo profound institutional changes that compromise their ability to fulfill their public interest missions. Applying this academic approach to SIDN's case, I identified several concerning changes that serve as warning signs for the potentially more serious consequences that could also befall hospitals, schools, and government agencies when they migrate to hyperscaler clouds.

In the case of SIDN, as outlined in my report, I highlight how these institutional changes might manifest if it proceeds with the move to AWS. Even before the move to AWS, we saw a shift at SIDN: technical expertise moved away from in-house capabilities toward vendor relationship management, with a focus on hiring AWS specialists over experts with broader technical skills. Likewise, there is a risk that SIDN's decision-making priorities might gradually align with AWS's business models rather than public interest mandates. SIDN's organizational culture, known for its technical tinkering and independent do-it-yourself attitude, will absorb the values and priorities of its cloud provider AWS. Importantly, these changes likely occur regardless of where .nl physically resides, or which national jurisdiction applies—they are inherent to the commercial cloud model itself and its profit-seeking imperatives.

An 'AWS sovereign cloud' is no way out of this conundrum. Nor is adding more of the same, but with a 'made in Europe' stamp. The Dutch academic community is particularly insightful in articulating these deeper concerns. Rather than fixating on reestablishing national control over infrastructure, academics at Utrecht and Amsterdam universities published open letters in March and April 2025, calling on their Executive Boards to "change course" and reduce hyperscaler cloud dependency. Their argument extends beyond conventional sovereignty and competition. Instead, they highlight how cloud migration converts universities "from being a source of technical innovation and knowledge distribution to consumers of services"—a fundamental shift that undermines academic independence and transforms knowledge-producing institutions into service-consuming clients. Or, in other words, they pursue better terms for the debate that address the harms arising from the cloud, beyond relying on shallow nationalistic terms and competition-oriented solutions.

David versus Cloud-liath

We must be clear-eyed about the challenges facing alternatives to hyperscaler clouds. The trillion-dollar valuations, global reach, and entrenchment of AWS, Microsoft, and Google in technical ecosystems make them incumbents that grassroots efforts will likely never fully displace. Their infrastructure, service offerings, and economies of scale cannot be matched by local providers alone, raising questions about whether replacement should even be the goal.

Equally concerning is how many alternative cloud initiatives are framed through problematic lenses of "digital sovereignty" or "strategic autonomy"—concepts rooted in nationalistic, militaristic thinking that risk further securitizing essential digital infrastructure. Such framing often leads to nationalistic approaches that can be equally harmful to public interest values. The challenge is not simply to build a "Dutch cloud" or "European cloud" as nationalistic alternatives, but to develop alternative visions for the role of computing in society that prioritize people over profit maximization.

Creating a digital future beyond nostalgia or net expansion requires building an inclusive economy that works for everyone. The most revolutionary alternatives won't aim to match hyperscalers at their own game. Instead, they will question whether we need so much computing in the first place, and who truly stands to benefit. Or in other words, emancipatory digital futures will only emerge when we transcend not just technological dependencies but also the limited political and economic imaginations that constrain how we conceive, fund, and build alternatives.

The growing recognition of cloud computing's massive energy consumption and resource demands adds an environmental dimension to this resistance. At the recent Paris AI Action Summit, over 130 organizations signed a joint statement calling for our computational futures to align with planetary boundaries. These kinds of appeals deserve at least as much attention—if not more—than the voices of 70 industry leaders, and a handful of politicians, advocating for a continuation of the current economic and computational status quo, albeit in a more "homegrown" form. Systems and economies built on principles of sufficiency, sustainability, equity, and human rights—rather than computational abundance and market expansion—offer a more promising path forward. Even if they can't rival the raw processing power of commercial clouds. Ultimately, the true success of cloud resistance may not lie in perfect technological alternatives, but in reshaping societies so that digital infrastructures serve humanity, rather than extract from it.

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